

Yoonsung Chung

Berkeley, CA | +1 (341) 253-3028 | yschung1105@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

University of California, Berkeley — Berkeley, CA

Aug 2025 – May 2027

M.S. in Civil & Environmental Engineering (Construction Systems)

Hongik University — Seoul, South Korea

Feb 2025

B.S. in Civil & Environmental Engineering

WORK AND LEADERSHIP EXPERIENCES

Construction Engineering Intern, Gordon N. Ball, Inc., Walnut Creek, CA

May 2026 – Aug 2026

- Built a cost estimating and forecasting system enabling PMs to change cost inputs and immediately view projected final costs and budget variances; tracked budget, committed cost, cost to date, and installed quantities by cost code.
- Prepared quantity takeoffs and cost estimates for heavy civil bids, including Caltrans SR-9 bridge replacements and ACFCWCD Line A-1 Capacity Improvements; mapped quantities to owner bid items.
- Performed quantity takeoffs for the Treasure Island project to support construction cost estimating.

External Research Assistant, Resilient Infrastructure Lab, Hongik University

Dec 2024 – Jun 2025

- Collaborated with public agencies and professional engineering organizations to investigate urban ground subsidence in Seoul

Squad Leader, 3rd Security Group, 17th Infantry Division, Republic of Korea Army, Incheon

Feb 2021 – Aug 2022

- Led a squad securing civil infrastructure near Incheon International Airport; assessed connectivity vulnerabilities in Yeongjong Island's two-bridge transportation network.

PROJECTS

Grimes Hall Digital Twin, UC Berkeley

Oct 2025 – Present

- Developed a web-based Grimes Hall BIM and reality-capture viewer with selectable components and linked element properties.
- Optimized BIM loading and rendering through compression, batching, and instancing; reduced draw calls from 49,285 to 244 in a controlled Fire-model test.
- Integrated COLMAP SfM and 3D Gaussian Splatting in a Python reconstruction workflow; implemented ICP experiments for Scan-BIM alignment.

Cal Construction IPD Competition Team (Architect), UC Berkeley

Aug 2025 – Feb 2026

- Reviewed drawings and resolved interdisciplinary coordination and constructability issues for the ASC Region 6 & 7 competition.

Analysis and Prediction of Ground Subsidence in Gangnam District, Seoul

Sep 2024 – Aug 2025

- Built an AnyLogic agent-based model with GIS to analyze interactions among sewer pipelines, street trees, and the subway network.

Design of the Second Airport Railway Bridge Using PSC Box Girders, Incheon

Mar 2024 – Jul 2024

- Developed a conceptual PSC box-girder design using MIDAS Civil, Xtract, and Response-2000; Excellence Award, KCI Capstone Design Competition

Digital Transformation Project for Construction Standards in South Korea

Apr 2023 – Nov 2023

- Converted construction standards into Mermaid flowcharts with the Ministry of Land, Infrastructure and Transport and KSCE.

RESEARCH EXPERIENCE

Undergraduate Research Assistant, Hongik University

Mar 2023 – Dec 2024

- Investigated displacement recovery of RC columns using Shape Memory Alloy bars in the plastic hinge region (Prof. Eun-Soo Choi).
- Developed a ground subsidence prediction model for Seoul's Gangnam District under Prof. Jeryang Park using agent-based modeling (ABM) in AnyLogic and GIS to analyze interactions among sewer pipelines, street trees, and subway infrastructure.

CONFERENCE PRESENTATIONS

European Geosciences Union (EGU), General Assembly, Vienna, Austria

May 2025

- Presented "Facility-Level Resilience Analysis of Urban Flood Defense Systems" at the poster session.

AWARDS & HONORS

Excellence Award, Capstone Design Competition, Korea Concrete Institute (KCI)

Nov 2024

Student Award, Presidential Citation, Hongik University

Aug 2024

Undergraduate Research Assistant & Academic Excellence Scholarships, Hongik University

2021 – 2024

Silver Award, 2024 Digital Poster Competition, Korean Society of Civil Engineers (KSCE)

Jun 2024

SKILL SET

Technical: Bluebeam, Revit, Navisworks, AutoCAD, MIDAS Civil, AnyLogic, QGIS, Mermaid, Python, Microsoft Excel

License: [Engineer Civil \(National Technical Qualification, HRD Korea\)](#)